Decision Tree

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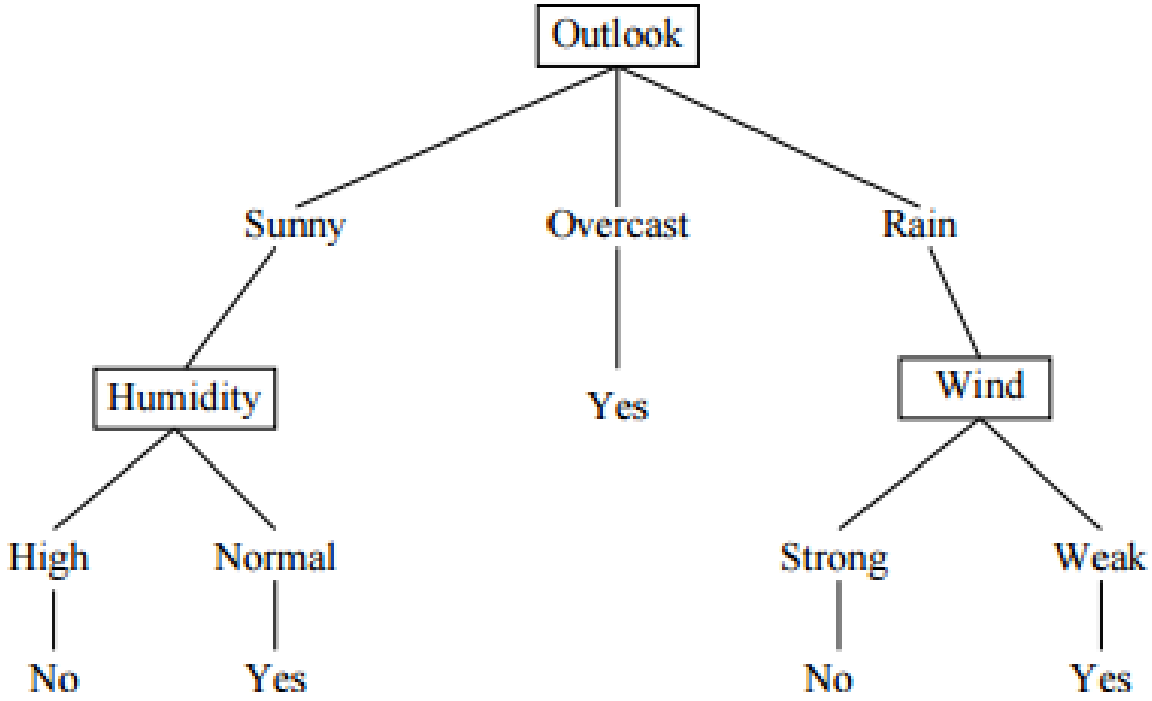
Decision Tree

- Machine Learning
  - Supervised Learning
    - Classification
    - Regression
  - Global Learning
  - Model-Based Learning
  - Eager Learning

Decision Tree

- **Decision tree** learning is a method for approximating **discrete-valued** and **real-valued target** functions, in which the learned function is represented as a decision tree
- Decision tree representation:
  - Each **internal node** tests an **attribute**
  - Each **branch** corresponds to **attribute value**
  - Each **leaf node** assigns a **classification**
- Re-representation as **if-then rules**: disjunction of conjunctions of constraints on the attribute value instances

Decision Tree for Play Tennis



Logical Formulation: (Outlook = Sunny ∧ Humidity = Normal)
∨ (Outlook = Overcast)
∨ (Outlook = Rain ∧ Wind = Weak)

When to Consider Decision Trees

- Instances describable by attribute–value pairs
- Target function is discrete and real valued
- Disjunctive hypothesis may be required
- Possibly noisy training data

Examples:

- Equipment or medical diagnosis
- Credit risk analysis
- Modelling calendar scheduling preferences

Top-Down Induction of Decision Trees (ID3)

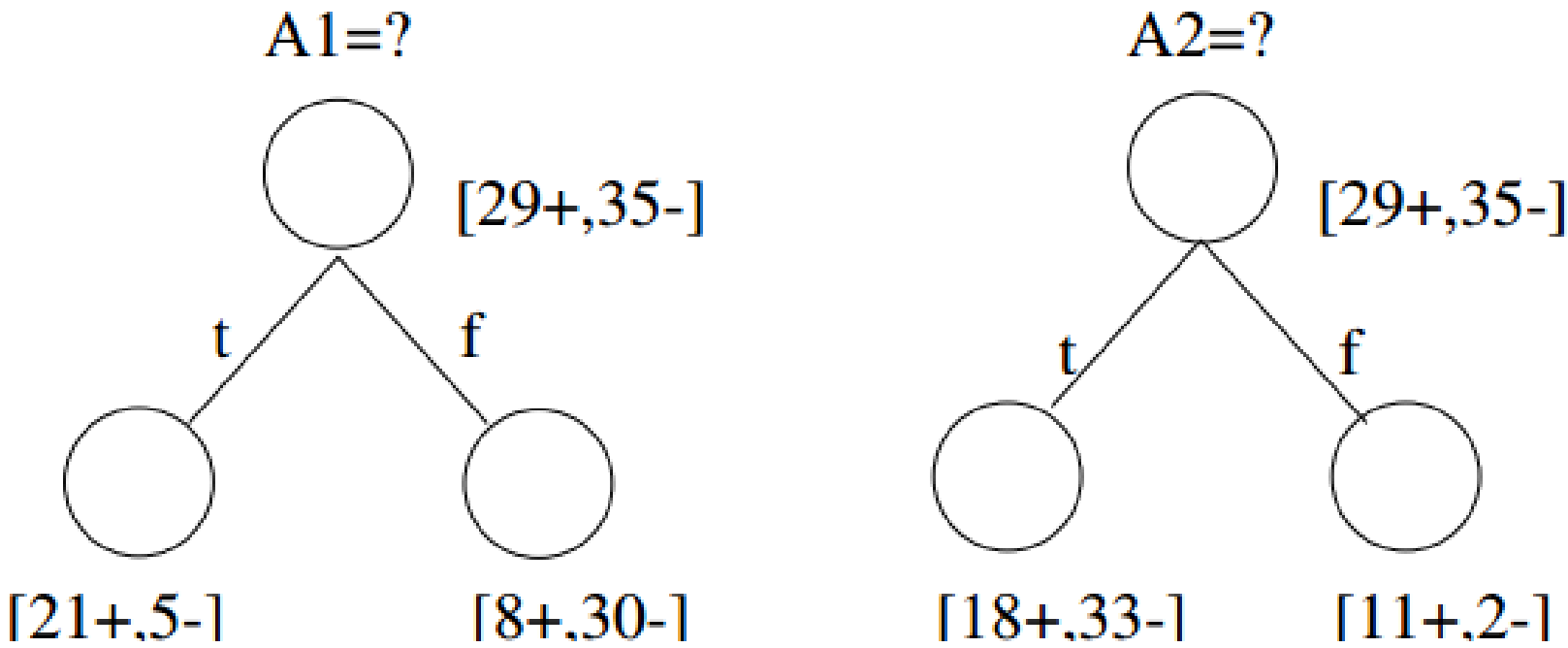
1. A is the “best” decision attribute for next node
2. Assign A as decision attribute for node
3. For each value of A, create new descendant of node
4. Sort training examples to leaf nodes
5. If training examples perfectly classified, Then STOP, Else iterate over new leaf nodes

Decision Tree Algorithms

Algorithms	ID3	C4.5	C5.0	CART
Type of data	Categorical	Continuous and Categorical	Continuous and Categorical, dates, times, timestamps	continuous and nominal attributes data
Speed	Low	Faster than ID3	Highest	Average
Pruning	No	Pre-pruning	Pre-pruning	Post pruning
Boosting	Not supported	Not supported	Supported	Supported
Missing Values	Can't deal with	Can't deal with	Can deal with	Can deal with
Formula	Use information entropy and information Gain	Use split info and gain ratio	Same as C4.5	Use Gini diversity index

Comparisons between different Decision Tree Algorithms

Which attribute is best?



Which attribute is best?

Play Tennis	Outlook	Temperature	…
Yes	Overcast	Hot	…
Yes	Overcast	Mild	…
Yes	Overcast	Cold	…
No	Sunny	Hot	…
No	Rain	Mild	…
No	Rain	Cold	…

Play Tennis	Outlook	Temperature
Yes	Overcast	Hot
Yes	Overcast	Mild
Yes	Overcast	Cold
No	Sunny	Hot
No	Rain	Mild
No	Rain	Cold

Play Tennis	Outlook	Temperature
Yes	Overcast	Hot
Yes	Overcast	Mild
Yes	Overcast	Cold
No	Sunny	Hot
No	Rain	Mild
No	Rain	Cold

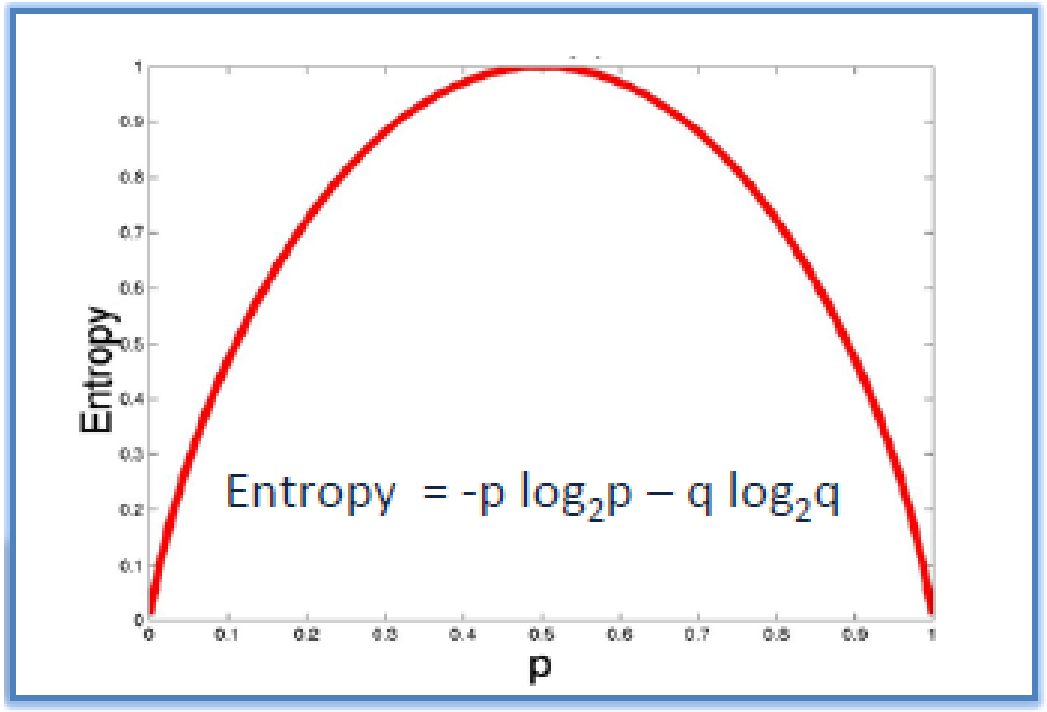
Entropy

- **S** is a sample of training examples
- **P+** is the proportion of **positive** examples in **S**
- **P-** is the proportion of **negative** examples in **S**
- **Entropy** measures the **impurity** or **uncertainty** of **S**

  $$Entropy(S) \equiv H(S) \equiv -p+ \log_2 p+ - p- \log_2 p-$$

- **H(S) = 0** if **sample is pure** (**all +** or **all -**), **H(S) = 1 bit** if **p+ = p- = 0.5**

Entropy: Coin Flipping





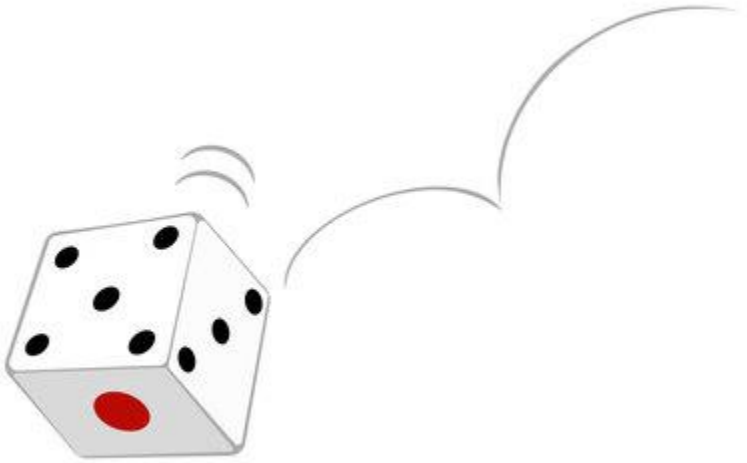
$$Entropy = -1 \times \log_2 1 - 0 \times \log_2 0 = 0$$

$$Entropy = -0.75 \times \log_2 0.75 - 0.25 \times \log_2 0.25 = 0.81$$

$$Entropy = -0.5 \times \log_2 0.5 - 0.5 \times \log_2 0.5 = 1$$

Entropy: Roll of the Dice





$$Entropy = -\frac{1}{6} \times \log_2 \frac{1}{6} - \frac{1}{6} \times \log_2 \frac{1}{6} - \frac{1}{6} \times \log_2 \frac{1}{6} - \frac{1}{6} \times \log_2 \frac{1}{6} - \frac{1}{6} \times \log_2 \frac{1}{6} - \frac{1}{6} \times \log_2 \frac{1}{6} = 2.58$$

Entropy

Number of Outputs (Classes)	Entropy (bits)	Minimum Number of Bits Required	Bit Values (Binary Combinations)
2	1	1	0, 1
3	1.585	2	00, 01, 10, 11
4	2	2	00, 01, 10, 11
5	2.322	3	000, 001, 010, 011, 100, 101, 110, 111
6	2.585	3	000, 001, 010, 011, 100, 101, 110, 111

With ***k*** bits could be represented $\boldsymbol{2^k}$ distinct outputs:

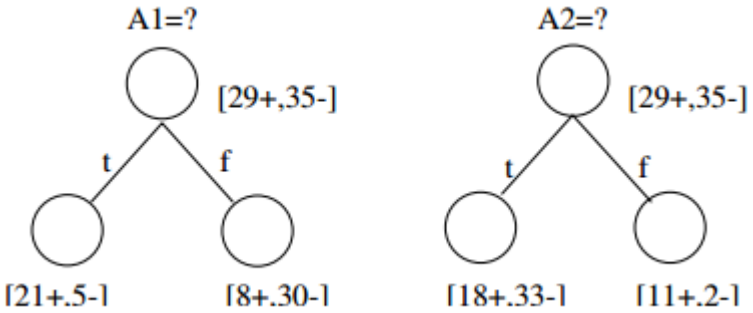
- **1** bit: $\mathbf{2^1 = 2}$ outputs (e.g., **0** and **1**).
- **2** bits: $\mathbf{2^2 = 4}$ outputs (e.g., **00**, **01**, **10**, **11**).
- **3** bits: $\mathbf{2^3 = 8}$ outputs (e.g., **000**, **001**, …, **111**).

Information Gain

- **Gain(S, A)** = expected reduction in entropy due to sorting on A

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

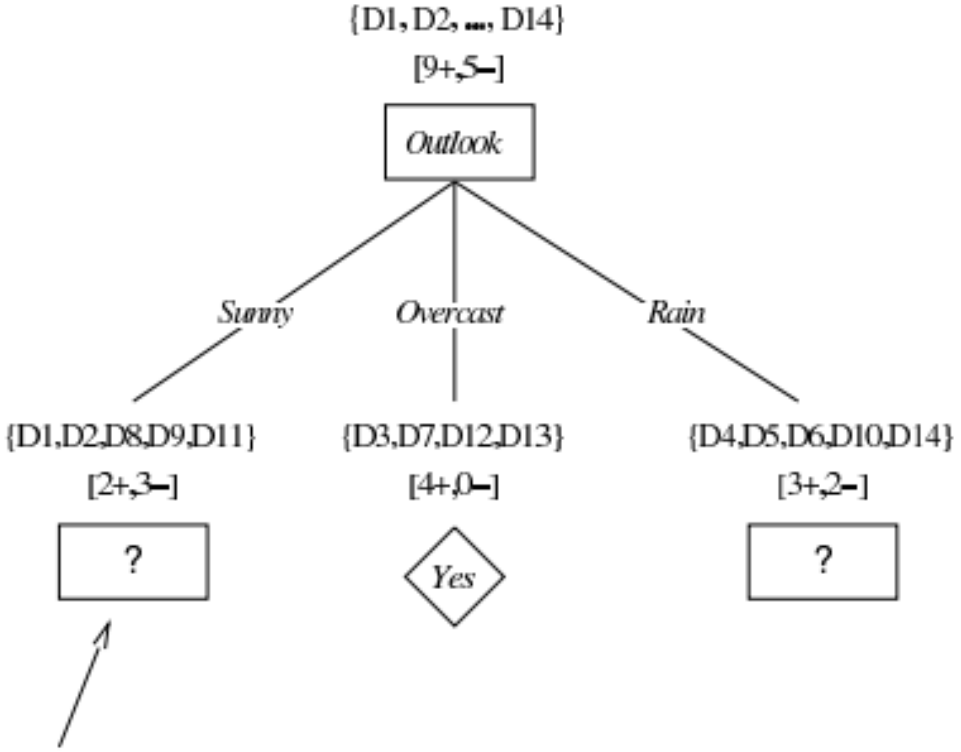
- **Information gain** is also called the **mutual information** between A and the labels of S



Training Examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Building the Decision Tree



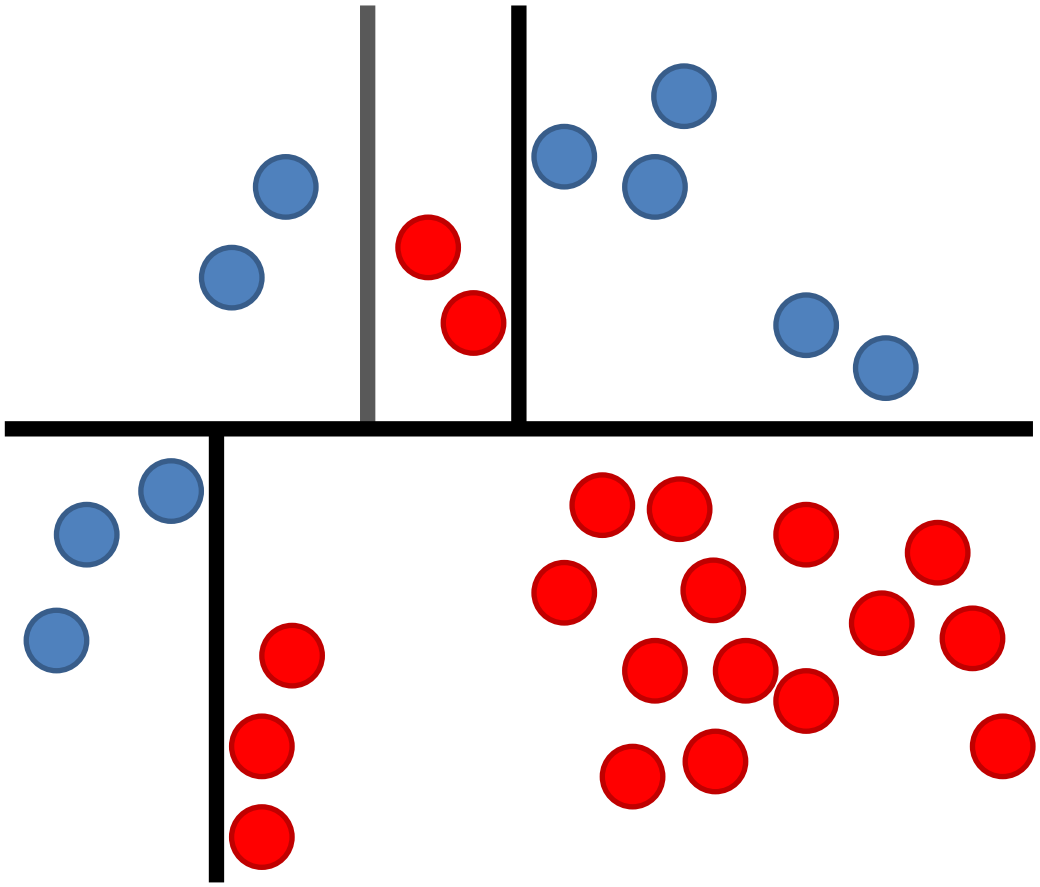
S_{sunny} = {D1,D2,D8,D9,D11}

Gain (S_{sunny}, *Humidity*) = .970 − (3/5) 0.0 − (2/5) 0.0 = .970

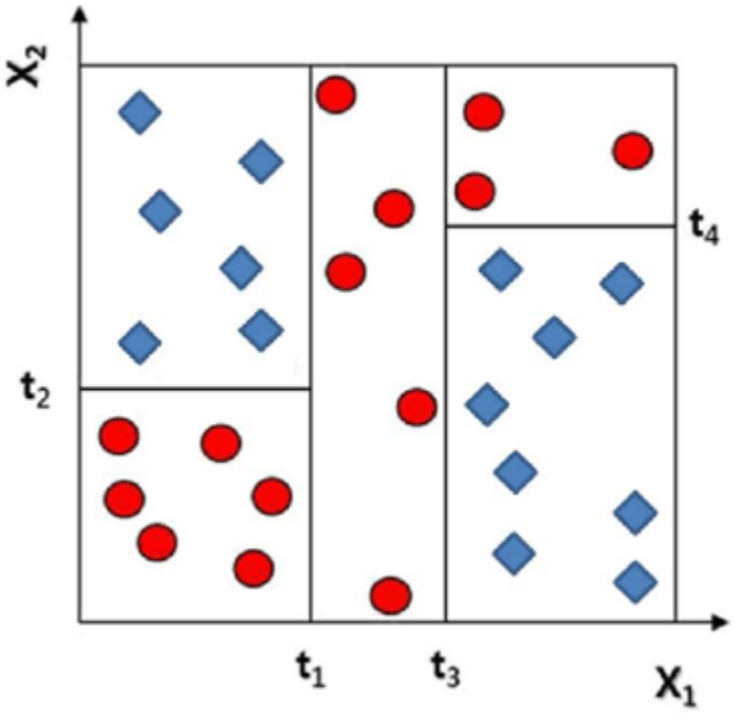
Gain (S_{sunny}, *Temperature*) = .970 − (2/5) 0.0 − (2/5) 1.0 − (1/5) 0.0 = .570

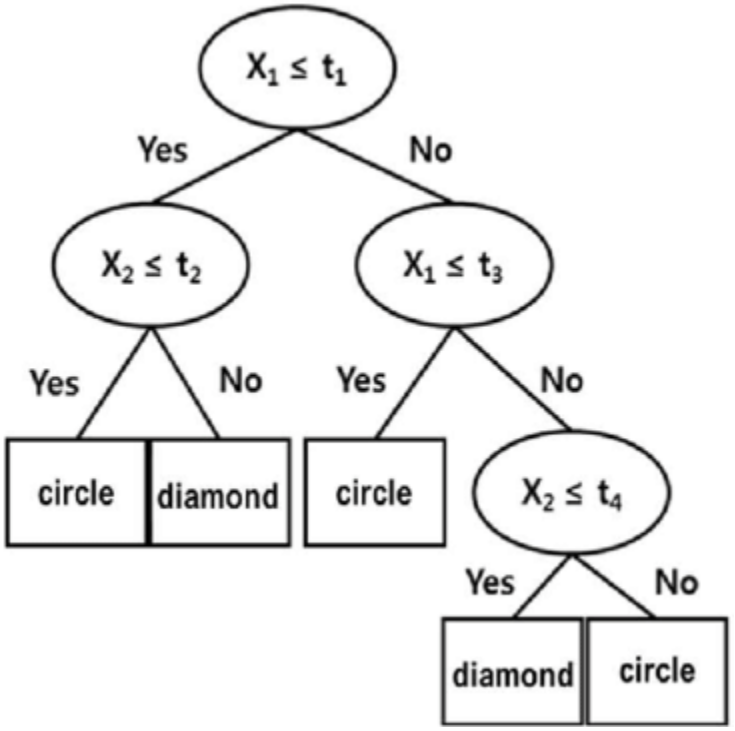
Gain (S_{sunny}, *Wind*) = .970 − (2/5) 1.0 − (3/5) .918 = .019

Decision Tree Visualization



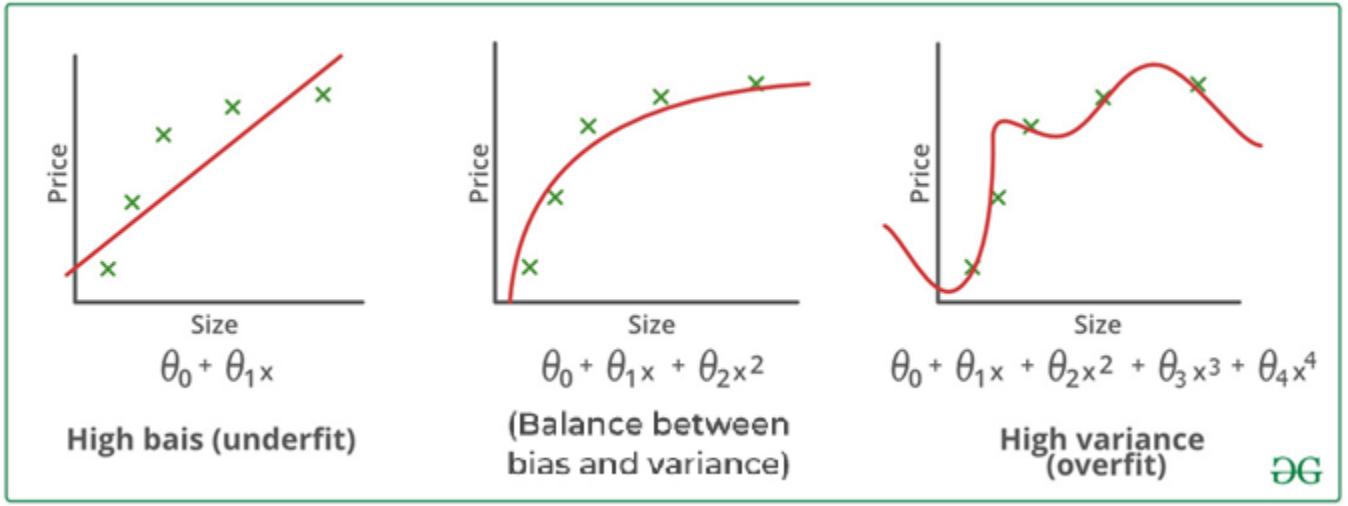
Decision Tree Visualization

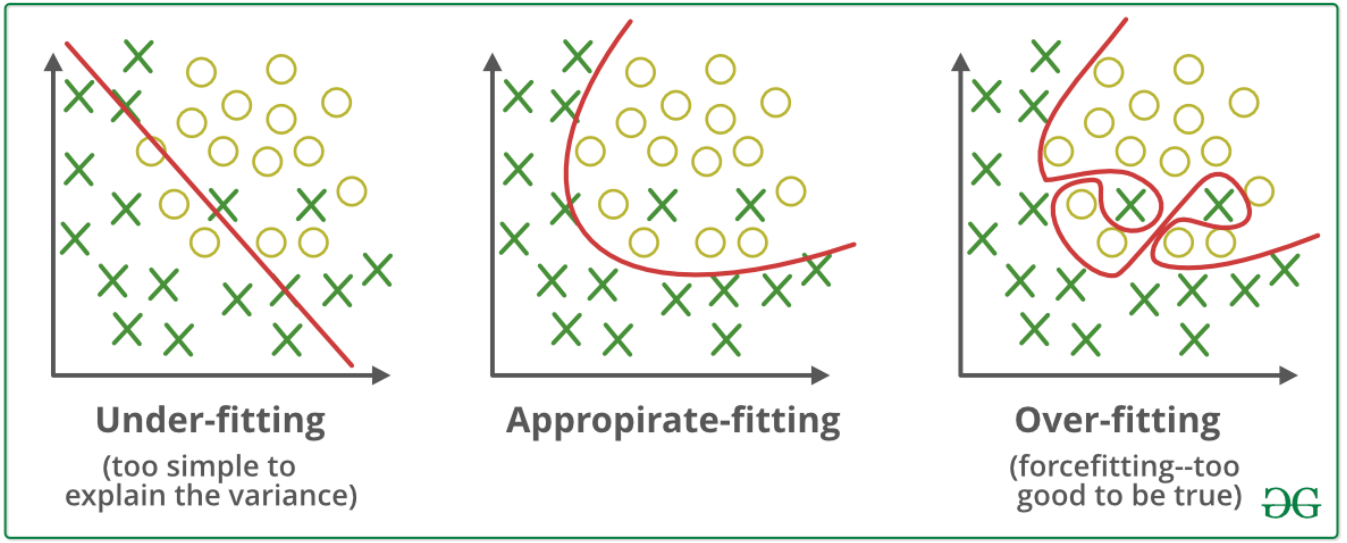




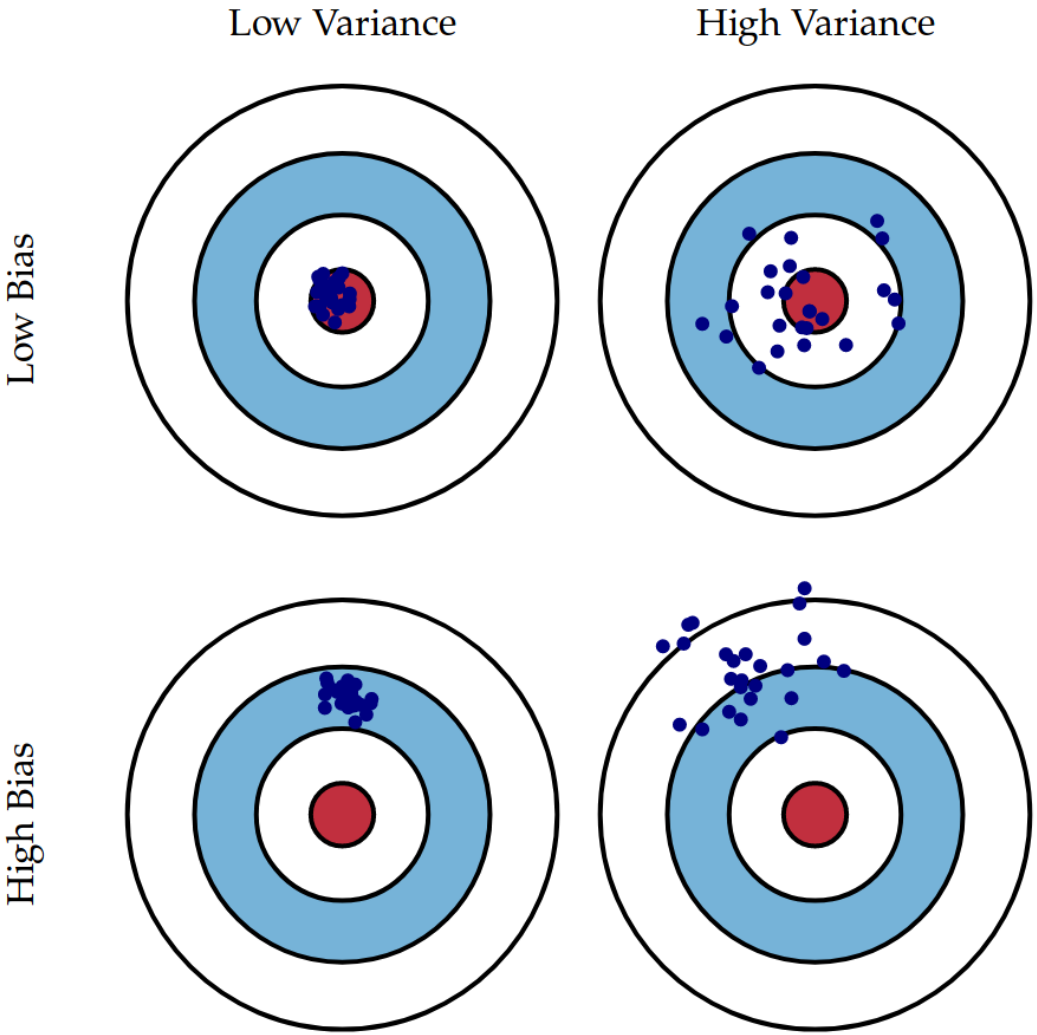
** Taken from [Improving Environmental Sustainability by Characterizing Spatial and Temporal Concentrations of Ozone]*

Overfitting vs Underfitting





Bias vs Variance



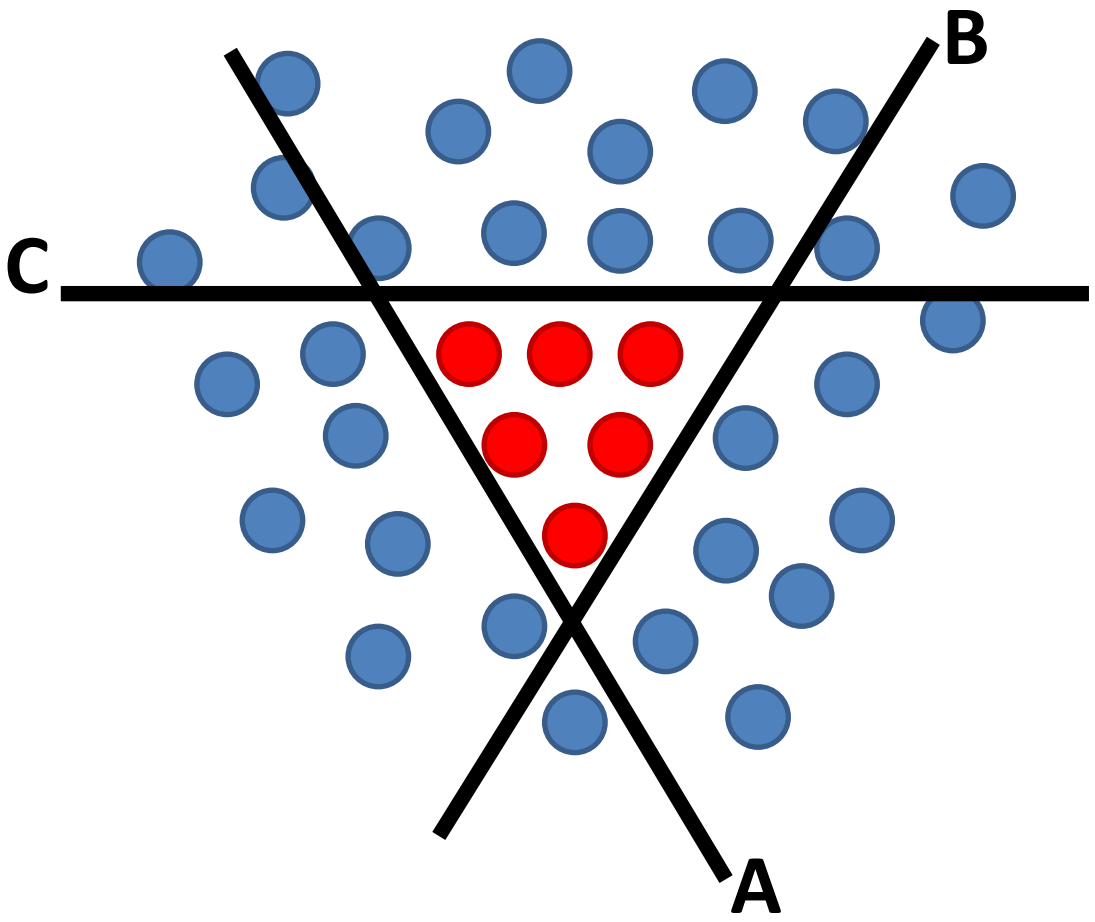
How to Fix the Overfitting?

- Common technics
  - Getting more training data
  - Cross-validation sampling
  - Reducing number of features
  - Pruning
  - Regularization
    - Increase Regularization term

How to Fix the Overfitting?

- Mostly for Decision Trees
  - Pruning
    - Pre-Pruning
    - Post-Pruning
  - Ensemble Learning
    - Random Forest
    - XGBoost

Ensemble Learning

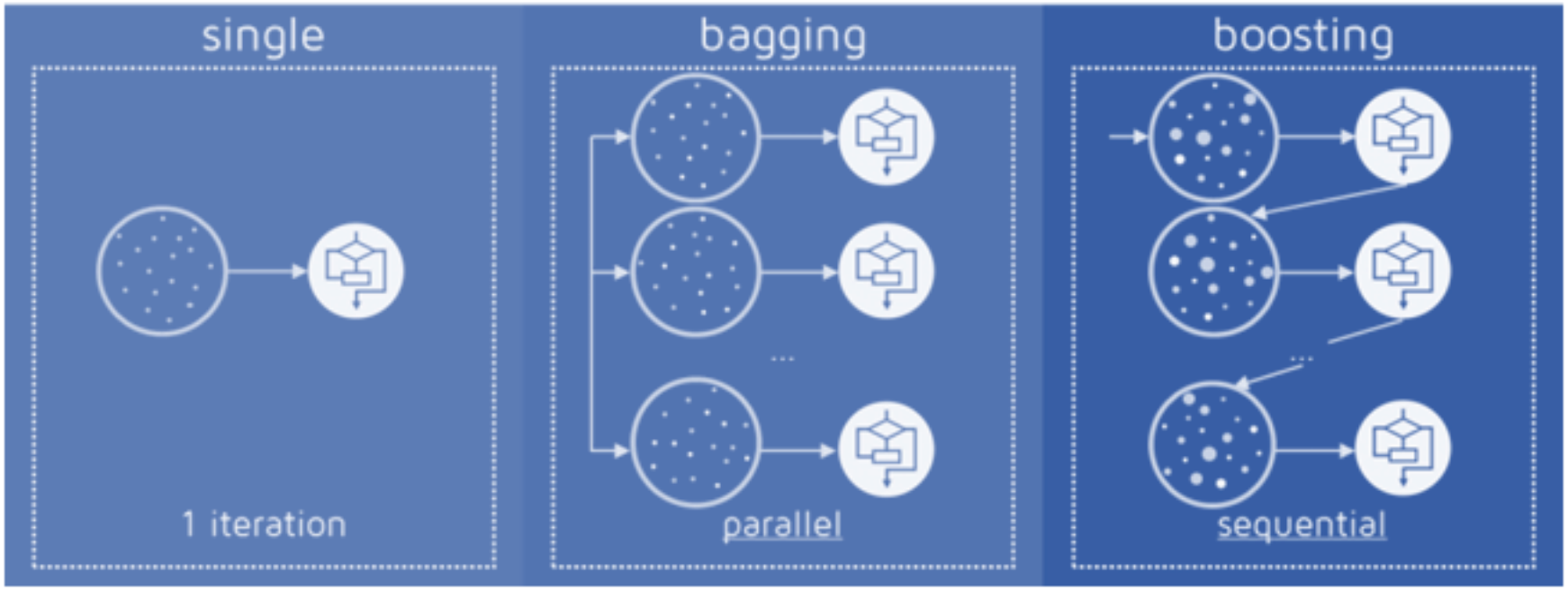


Ensemble Learning

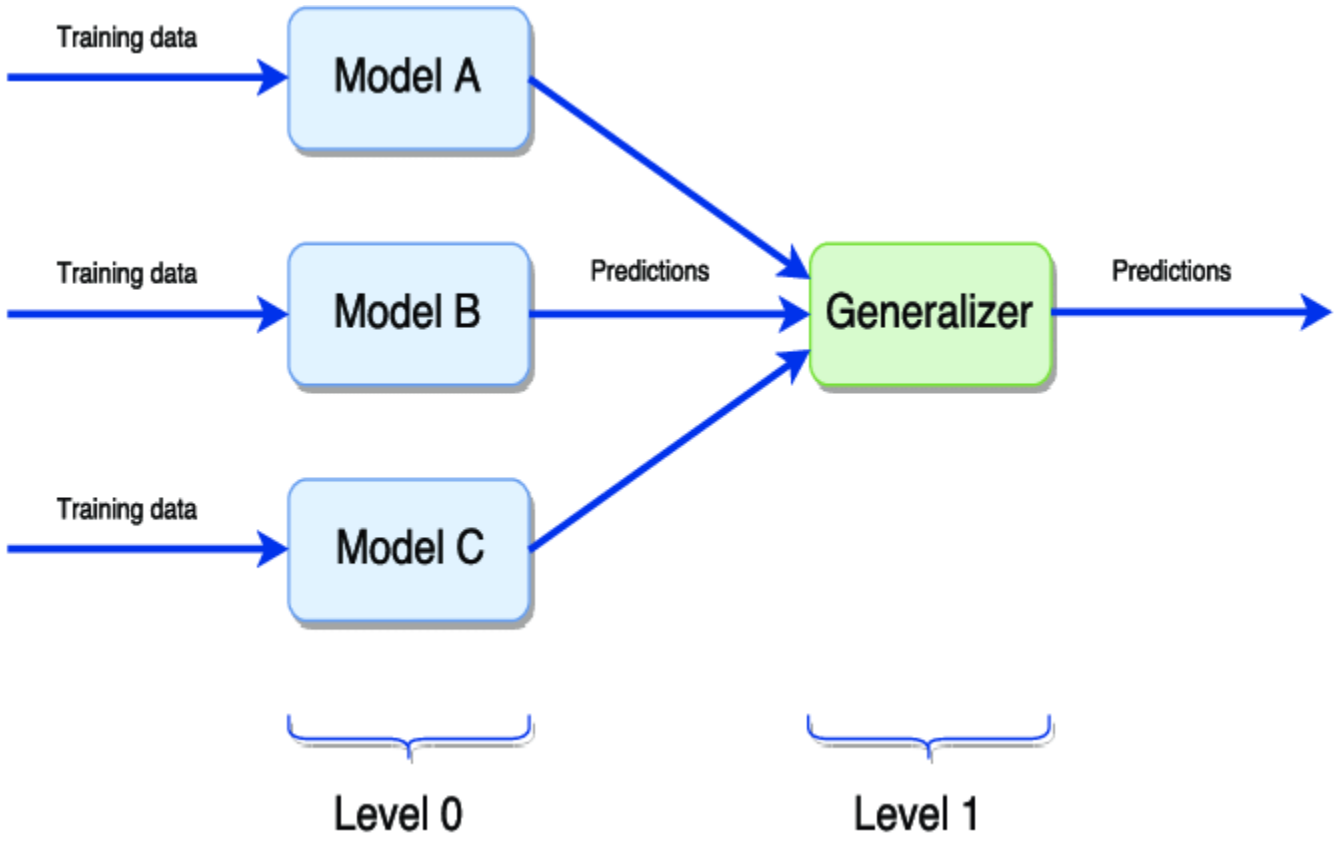
- Bagging (Bootstrap aggregating)
- Boosting
- Stacking

* *https://en.wikipedia.org/wiki/Bootstrap_aggregating*
* *https://quantdare.com/what-is-the-difference-between-bagging-and-boosting/*

Bagging vs Boosting



Stacking Ensemble Learning



Example

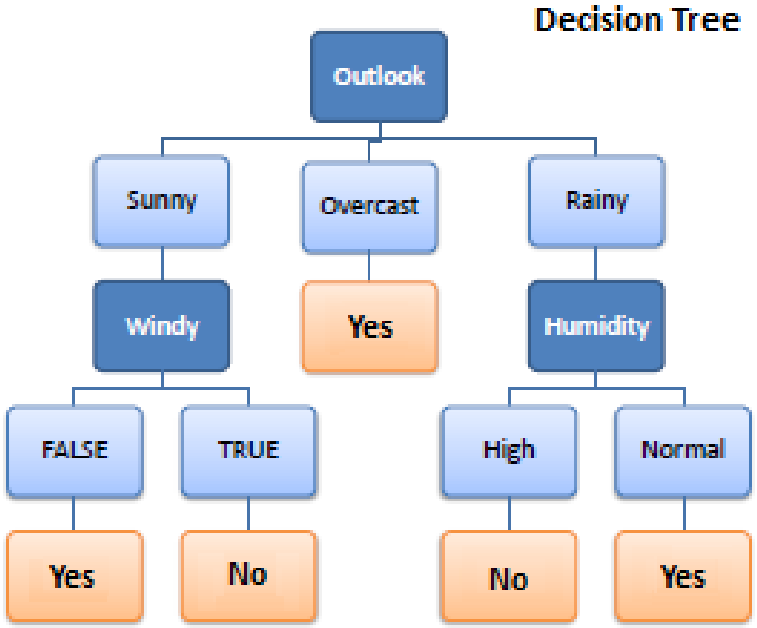
https://www.saedsayad.com/decision_tree.htm

Dataset & Decision Tree

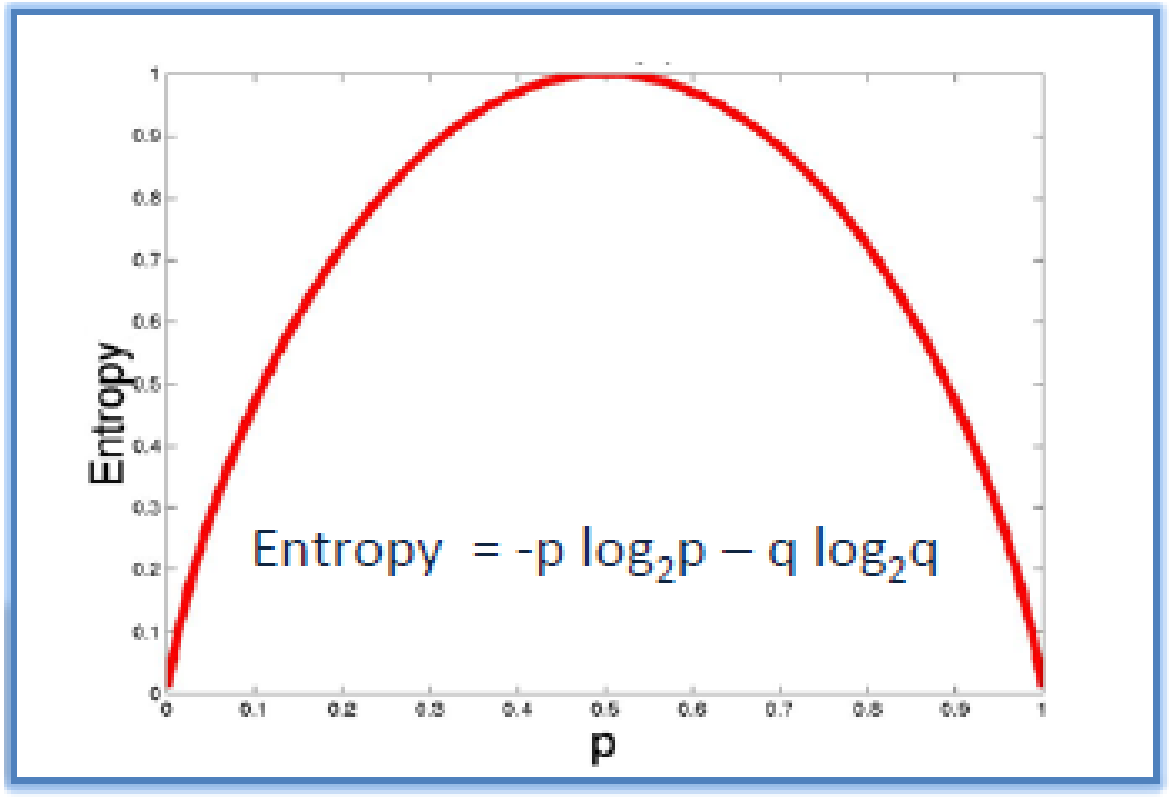
Predictors | Target

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No





Entropy for 2 Outcomes: $\boldsymbol{p}$ and $\boldsymbol{q}$



Entropy = $-0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1$

Entropy for One Attribute

$$E(S) = \sum_{i=1}^{c} - p_i \log_2 p_i$$

Play Golf	
Yes	No
9	5



Entropy(PlayGolf) = Entropy (5,9)

= Entropy (0.36, 0.64)

= - (0.36 $\log_2$ 0.36) - (0.64 $\log_2$ 0.64)

= 0.94

Entropy for Two Attributes

$$E(T,X) = \sum_{c \in X} P(c)E(c)$$

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14



E(PlayGolf, Outlook) = **P**(Sunny)***E**(3,2) + **P**(Overcast)***E**(4,0) + **P**(Rainy)***E**(2,3)

= (5/14)*0.971 + (4/14)*0.0 + (5/14)*0.971

= 0.693

Information Gain *(for Outlook)*

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

G(PlayGolf, Outlook) = **E**(PlayGolf) – **E**(PlayGolf, Outlook)

= 0.940 – 0.693 = 0.247

Information Gain

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			

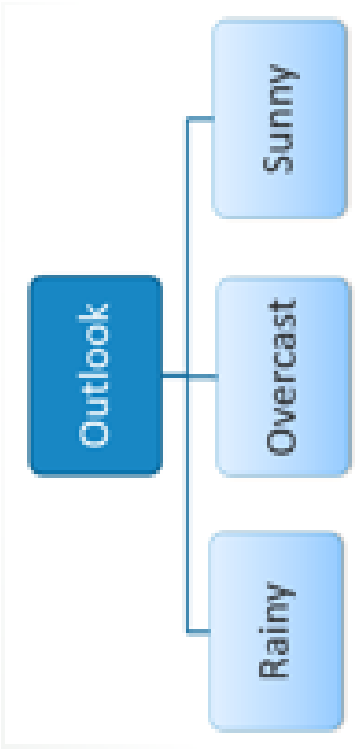
		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1
Gain = 0.029			

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1
Gain = 0.152			

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3
Gain = 0.048			

The Best Attribute is the One with the Highest Information Gain

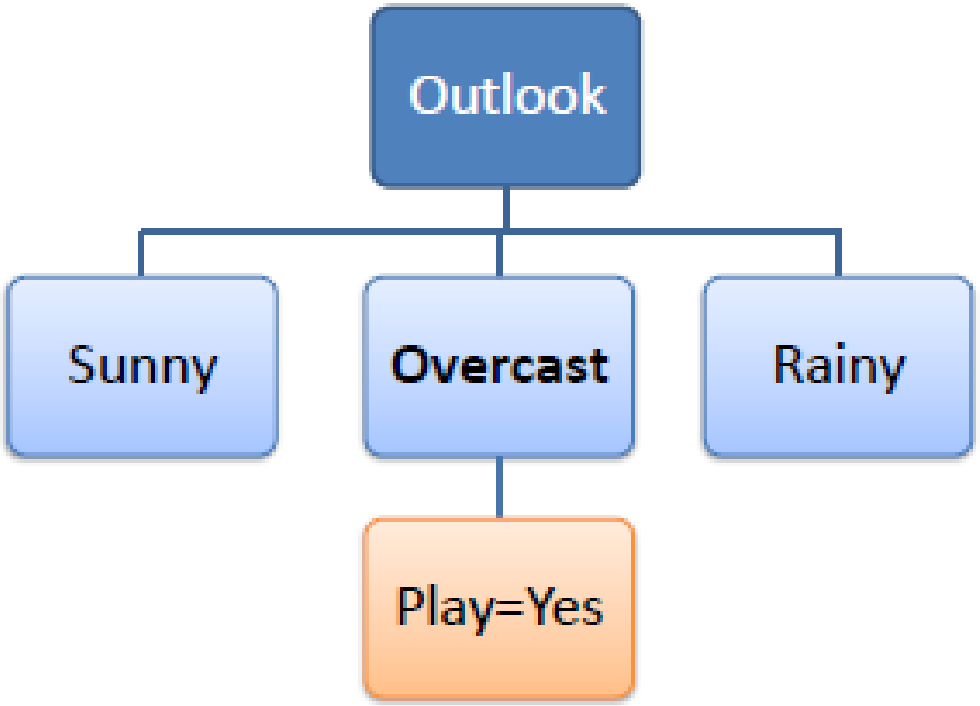
★		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			



Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Overcast	Cool	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes

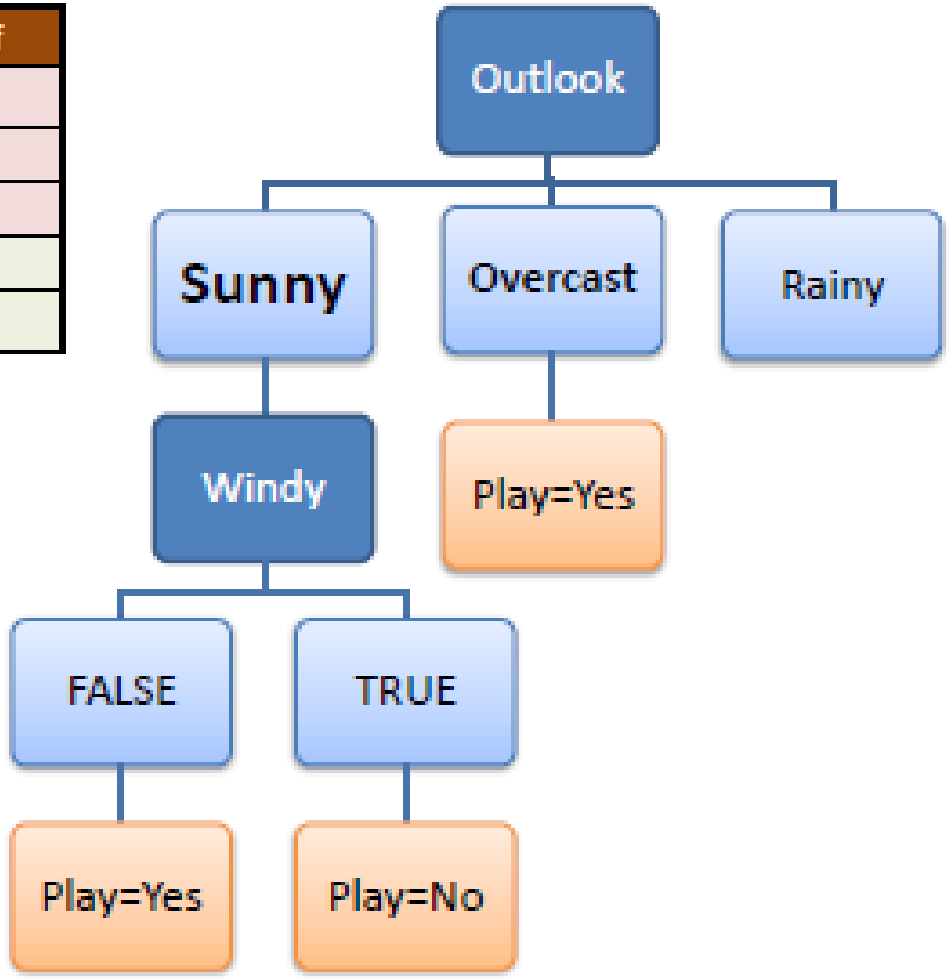
Branch with Entropy of 0

Temp.	Humidity	Windy	Play Golf
Hot	High	FALSE	Yes
Cool	Normal	TRUE	Yes
Mild	High	TRUE	Yes
Hot	Normal	FALSE	Yes



Branch with Entropy More than 0

Temp.	Humidity	Windy	Play Golf
Mild	High	FALSE	Yes
Cool	Normal	FALSE	Yes
Mild	Normal	FALSE	Yes
Cool	Normal	TRUE	No
Mild	High	TRUE	No



Decision Rules & Decision Tree

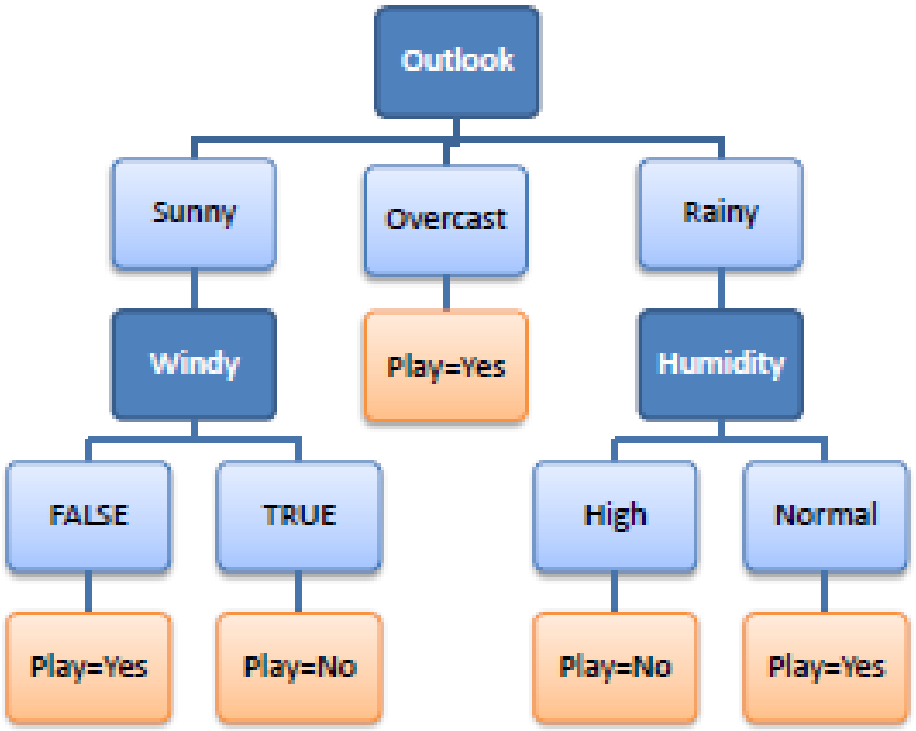
R_1: IF (Outlook=Sunny) AND (Windy=FALSE) THEN Play=Yes

R_2: IF (Outlook=Sunny) AND (Windy=TRUE) THEN Play=No

R_3: IF (Outlook=Overcast) THEN Play=Yes

R_4: IF (Outlook=Rainy) AND (Humidity=High) THEN Play=No

R_5: IF (Outlook=Rain) AND (Humidity=Normal) THEN Play=Yes



Homework: ID3

Implement a ID3 (Decision Tree) that classifies breast cancer using the following data:

- https://archive.ics.uci.edu/ml/datasets/breast+cancer

Addition to the task:

- To test the algorithm, apply 10-fold cross-validation (divide the data randomly into 10 sets and do 10 trainings and tests of the model, each time using one of the sets for testing and the other 9 for training).
- Display the Accuracy metric for each of the 10 trainings, as well as their arithmetic mean (for a summary assessment of the classifier).

** To avoid overfitting of the tree, use a constant K defining the minimum number of training examples in the dataset.*